using these equations and ideas, to prove my claims, in Chapter 27, about the *Endurance*'s trajectory, the trajectory's instability on the rim of the volcano, and its launch toward Edmunds' planet.

Chapter 30. Messaging the Past

In the bulk as well as in our brane, the locations in spacetime, to which messages and other things can travel, are controlled by the law that nothing can travel faster than light. We physicists use spacetime diagrams to explore the consequences of this law. We draw spacetime diagrams in which, at each event, there is a "future light cone." Light travels outward from that event along the light cone; everything else, moving slower than light, travels from that event either along or inside the cone. See, for example, *Gravity: An Introduction to Einstein's General Relativity* (Hartle 2003).

Figure TN.1 shows the pattern of future light cones inside and on faces of the tesseract, in my interpretation of *Interstellar*. (It is the mathematical description of spacetime warping that I refer to in footnote 1 of Chapter 30. Physicists call this pattern of light cones "the causal structure of spacetime" inside the tesseract.) Figure TN.1 also shows the world line (violet curve) of the gravitational-wave message (force) sent by Cooper through the tesseract's interior to Murph's bedroom; and the world line (red dashed line) of the light ray from the bedroom through tesseract faces, by which Cooper sees the bedroom. This is a spacetime version of the purely spatial diagram in Figure 30.5.